Neighborhood Label-Barycenter LMB Fusion for Distributed Multi-Target Tracking Under Unreliable Communication

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Abstract— Packet loss in a peer-to-peer labeled multi-Bernoulli tracking network creates a component-correspondence problem: neighboring sensors can keep different labels for the same physical target, or keep the same label on states that have drifted apart, so average fusion can mix posterior components that are not label-aligned. This paper develops a neighborhood label-barycenter operator for distributed LMB fusion: each sensor chooses a median-cardinality medoid reference from its communication neighborhood, matches neighboring tracks to the reference by assignment, and replaces matched posteriors by a first-two-moment barycenter over the matched group. The construction separates label canonicalization from posterior averaging, which makes it possible to test whether performance gains come from label-set agreement alone or from matched spatial barycenters. In a 50-trial tiered packet-loss formation scenario, the proposed operator reduces network OSPA disagreement by 81.59%, local E-OSPA by 17.15%, and local RMSE by 6.35% relative to a tuned spatial-KLA AA baseline, while a reference-only ablation gives only 0.54% RMSE reduction.

Index Terms— Arithmetic average fusion, distributed sensor networks, labeled multi-Bernoulli filter, multi-target tracking, random finite sets.

I. INTRODUCTION

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This manuscript uses only simulated multi-target tracking data. No human or animal subjects are involved.

Code and reproducibility artifacts are available at [repository DOI/URL].

DISTRIBUTED multi-target tracking has to solve two coupled problems at once: estimating an unknown and time-varying set of target states, and keeping the local estimates at different sensors mutually usable when communication is lossy. Labeled random finite sets (RFSs) and the labeled multi-Bernoulli filter provide a principled representation for target birth, persistence, death, and track identity [1], [2]. In a centralized or well-synchronized system, labels make posterior components interpretable as tracks. In a peer-to-peer sensor network, however, labels also encode local filtering history. After asymmetric measurements and packet drops, two sensors can keep different labels for the same physical target, or keep the same label for track states that have drifted apart.

Density fusion for distributed RFS tracking is usually built around conservative geometric rules, such as generalized covariance intersection or the Kullback–Leibler average (KLA), or around arithmetic-average (AA) fusion rules that preserve weak components and support heterogeneous labeled and unlabeled filters [3]–[6]. These rules define how much probability mass each neighbor should contribute. They do not, by themselves, decide which Bernoulli component in one local LMB posterior refers to which component in another. Once unmatched components enter the existence and spatial fusion consumers, better scalar weights can attenuate a bad contribution but cannot create a missing component correspondence.

This paper starts from that structural failure mode. We use a tuned AA-LMB baseline in which fusion weights are resolved at the object level before the existence and spatial consumers, so the comparison is not confounded by scalar weight routing. The remaining errors separate into two questions: which local label set should serve as the neighborhood reference, and how should the matched posterior states be replaced after correspondence is established? This separation turns the design into a testable label canonicalization and posterior barycenter problem, rather than another scalar weight-search problem.

We propose a neighborhood label-barycenter operator for distributed LMB fusion. At each time step, each sensor uses only the estimates available in its communication neighborhood. The sensor first chooses a reference estimate by a median-cardinality medoid rule, which prevents minority cardinality outliers from directly selecting the reference. It then matches neighboring estimates to the reference labels using the Hungarian assignment method [7]. For each reference label, the matched Gaussian states are replaced by the mean and covariance of the empirical matched mixture. Repeating this local operator over the graph propagates label-space agreement without reading a global label set.

The contribution is threefold.

- We diagnose a component-correspondence failure in AA-LMB fusion under unreliable communication: label-set disagreement and matched-state dispersion remain even after target-wise weight handling is corrected.
- We introduce a graph-local label-barycenter operator that combines median-cardinality medoid reference selection, assignment-based label canonicalization, and matched first-two-moment posterior barycenters.
- We give structural properties and paired ablations that separate label-set copying from posterior barycentering, showing that barycentering is responsible for the main spatial tracking gain.

II. RELATED WORK

Labeled RFS filters provide a Bayesian finite-set formulation for preserving target identity while estimating multi-object states [1]. The LMB filter gives a tractable approximation in which each Bernoulli component carries an existence probability and a labeled single-target density [2]. Multi-sensor and distributed versions of labeled RFS filters have been studied through generalized covariance-intersection fusion, robust distributed labeled-RFS fusion, consensus LMB updates, and resilient peer-to-peer LMB fusion [4], [5], [8], [9]. These methods address how sensors should share uncertain multi-object information, but practical labeled fusion still depends on whether components with different local labels are comparable.

KLA and logarithmic opinion-pool fusion are widely used because they avoid double counting under unknown correlations and connect naturally to consensus on probability densities [3]. This conservatism is valuable in distributed tracking, but geometric fusion can suppress weak components when local support is asymmetric. Arithmetic-average fusion follows a linear-pooling principle and has been analyzed statistically and information-theoretically for RFS densities [10]. Subsequent AA density-fusion work gives unified derivations for unlabeled and labeled RFS fusion, heterogeneous unlabeled/labeled fusion, and variational heterogeneous distributed fusion [6], [11], [12]. Fisher-information weighted AA fusion for multi-rate filters and weighted peer-to-peer RFS fusion further show that recent work is moving from one-shot density pooling toward communication- and filter-quality-aware AA designs [13], [14].

Heterogeneity and field-of-view mismatch create a separate difficulty. RFS fusion across heterogeneous sensors can be formulated without assuming identical local multi-object densities [15], and labeled RFS fusion with different fields of view has been studied explicitly [16]. Information-geometric labeled-RFS fusion offers another route for distributed heterogeneous state estimation [17]. These lines motivate robust fusion under nonidentical local views, but they do not by themselves remove the output-level failure in which two local LMB filters carry different labels for the same physical target after packet loss.

Our method is complementary to AA, KLA, and information-geometric weighting. It does not replace the underlying density-fusion rule. Instead, it supplies a graph-local label-aligned projection layer that makes local Bernoulli components comparable before the fused output is evaluated. Finite-set fusion also raises consistency questions that are not present in single-target Gaussian fusion: pointwise consistency and global cardinality can move in different directions [18]. We therefore report both network disagreement and truth-referenced tracking error using OSPA-style metrics [19]. The method relies on assignment, for which the Hungarian method remains the standard polynomial-time solver [7], and on graph-local iteration, which is closely related to average-consensus ideas in sensor networks [20].

III. PROBLEM FORMULATION

Consider a network of S sensors. At time k, sensor s outputs an LMB estimate

$$X_s(k) = \{(\ell_{s,i}, r_{s,i}, \mu_{s,i}, \Sigma_{s,i})\}_{i=1}^{n_s(k)}, \tag{1}$$

where $\ell_{s,i}$ is a label, $r_{s,i} \in [0,1]$ is existence probability, and $(\mu_{s,i}, \Sigma_{s,i})$ are the first two moments of a Gaussian state approximation. Let $\mathcal{N}_s$ denote the sensor indices available to sensor s through the communication graph, including s itself.

The core difficulty is that the sets $\{\ell_{s,i}\}$ need not agree across sensors. A local filter may keep a label after a missed packet while a neighboring filter deletes it, or two sensors may assign different labels to the same physical target. Direct AA or KLA fusion of unmatched Bernoulli components then mixes component identities and states. We seek an estimate-level operator

$$\mathcal{B}_s : \{X_j(k) : j \in \mathcal{N}_s\} \mapsto Y_s(k) \tag{2}$$

that is local to the graph, preserves an interpretable reference label set, and reduces both pairwise output disagreement and truth-referenced tracking error.

IV. NEIGHBORHOOD LABEL-BARYCENTER FUSION

A. Reference Selection

For each sensor s and time k, the candidate neighborhood is $\mathcal{N}_s$. Let $n_j(k)$ be the number of tracks output by neighbor j. The reference cardinality is chosen around the sample median,

$$m_s(k) = \mathrm{median}\{n_j(k) : j \in \mathcal{N}_s\}. \tag{3}$$

The candidate set is

$$\mathcal{C}_s(k) = \arg\min_{j \in \mathcal{N}_s} |n_j(k) - m_s(k)|. \tag{4}$$

Among these candidates, the reference is the medoid under an OSPA-like set distance d_c with cutoff c:

$$\rho_s(k) = \arg \min_{j \in \mathcal{C}_s(k)} \frac{1}{|\mathcal{N}_s|} \sum_{q \in \mathcal{N}_s} d_c(X_j(k), X_q(k)). \quad (5)$$

The labels of $X_{\rho_s(k)}(k)$ form the local reference label set. In the implementation, d_c is computed from position states by Hungarian matching, squared matched distances, and a cutoff penalty c^2 for unmatched tracks, normalized by the larger cardinality. The cutoff is the configured consensus cutoff when available and otherwise follows the OSPA cutoff used by the tracking evaluation.

B. Assignment-Based Label Canonicalization

Let the reference have states $\{\mu_{\rho,a}\}_{a=1}^{m}$. For each neighbor j, we construct a cost matrix

$$C_{ba} = \|p(\mu_{j,b}) - p(\mu_{\rho,a})\|_2, \quad (6)$$

where $p(\cdot)$ extracts position coordinates. The Hungarian method returns the minimum-cost assignment between neighbor states and reference states [7]. This assignment defines the matched group $\mathcal{G}_{s,a}(k)$ for each reference label a.

C. Matched Moment Barycenter

For a matched group $\mathcal{G}_{s,a}(k) = \{(\mu_i, \Sigma_i)\}_{i=1}^{M}$, the output state is

$$\bar{\mu}_{s,a} = \frac{1}{M} \sum_{i=1}^{M} \mu_i, \quad (7)$$

$$\bar{\Sigma}_{s,a} = \frac{1}{M} \sum_{i=1}^{M} \left[\Sigma_i + (\mu_i - \bar{\mu}_{s,a})(\mu_i - \bar{\mu}_{s,a})^T\right]. \quad (8)$$

The output $Y_s(k)$ keeps the reference labels and replaces their Gaussian moments by $(\bar{\mu}_{s,a}, \bar{\Sigma}_{s,a})$. The implementation repeats this neighborhood operator for $H = 3$ rounds in the reported experiments.

D. Graph Locality and Computational Cost

The operator is graph-local at every round: sensor s only reads estimates indexed by $\mathcal{N}_s$, and no global label dictionary is constructed. If $d_s = |\mathcal{N}_s|$ and $n_{\max}$ upper bounds the number of tracks in the neighborhood, the dominant cost is assignment. The medoid stage evaluates set distances among neighborhood estimates, while the matching stage solves at most d_s rectangular assignments to the chosen reference. With a cubic Hungarian implementation, the per-round cost is bounded by $O(d_s^2 n_{\max}^3 + d_s n_{\max}^3)$ in the worst case, plus linear moment accumulation over matched tracks. This cost is acceptable in the reported formation setting, but it explains why the method is slower than scalar-weight AA and motivates the runtime study reported with the experimental results.

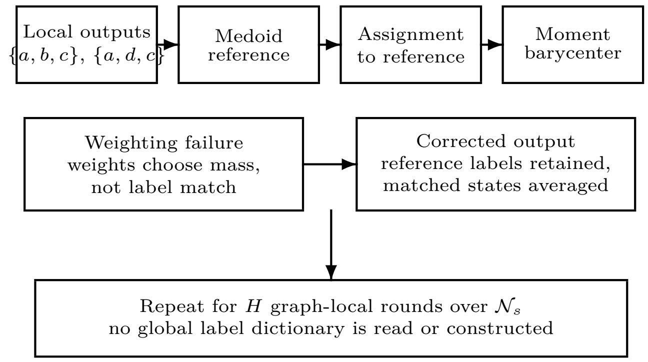

Fig. 1. Core mechanism of the proposed operator. Fusion weights specify how much probability mass a neighbor contributes, but they do not define component correspondence across local LMB posteriors. The operator therefore chooses a graph-local reference, matches neighbor tracks to that reference, and replaces matched states by posterior moment barycenters.

Inputs: neighborhood estimates $\{X_j(k) : j \in \mathcal{N}_s\}$, cutoff c, rounds H. Round h at sensor s: (i) select the median-cardinality candidate set $\mathcal{C}_s(k)$; (ii) choose the OSPA-medoid reference $\rho_s(k)$; (iii) assign neighbor states to reference states by Hungarian matching; (iv) moment-match each reference-label group; (v) write the label-aligned estimate $Y_s(k)$ as the next-round output.

Fig. 2. Implementation outline used in the validation. The operator is an estimate-level projection layer: it consumes only graph-neighborhood outputs and does not read a global label set.

V. STRUCTURAL PROPERTIES

Proposition 1 (Median-cardinality robustness):
If more than half of the sensors in $\mathcal{N}_s$ output the same cardinality $n^\star$ at time k, then the reference selected by the median-cardinality stage also has cardinality $n^\star$.

Proof:
With a strict majority at $n^\star$, the sample median of neighborhood cardinalities is $n^\star$. The minimum value of $|n_j(k) - m_s(k)|$ is therefore zero, and it is attained only by sensors with $n_j(k) = n^\star$. The subsequent medoid step is restricted to this candidate set. ■

Proposition 2 (Moment projection):
For a matched group $\mathcal{G} = \{(\mu_i, \Sigma_i)\}_{i=1}^{M}$, the barycenter mean minimizes $\sum_i \|z - \mu_i\|_2^2$, and $(\bar{\mu}, \bar{\Sigma})$ are the first two moments of the equally weighted Gaussian mixture $M^{-1} \sum_i \mathcal{N}(\mu_i, \Sigma_i)$.

Proof:
The least-squares objective has gradient $2Mz - 2\sum_i \mu_i$, hence its unique minimizer is $M^{-1} \sum_i \mu_i$. The

covariance identity follows from total covariance:

$$\mathrm{Cov}(x) = \mathbb{E}_i[\mathrm{Cov}(x|i)] + \mathrm{Cov}_i(\mathbb{E}[x|i]). \tag{9}$$

■

Proposition 3 (Stable-matching consensus limit):
For one matched reference label, suppose the correspondence is fixed and correct over a connected graph, and let

$$q_s^h = \left[r_s^h, (\mu_s^h)^T, \mathrm{vec}(M_s^h)^T\right]^T, \quad M_s^h = \Sigma_s^h + \mu_s^h(\mu_s^h)^T. \tag{10}$$

If $q_s^{h+1} = \sum_j W_{sj} q_j^h$ with a primitive doubly stochastic matrix W respecting the graph, then all q_s^h converge to $S^{-1}\sum_j q_j^0$. The limiting mean and second moment therefore equal the centralized equal-weight moment barycenter.

Proof:
Stack the q_s^h vectors. Since $W^h \to S^{-1}\mathbf{1}\mathbf{1}^T$, every node converges to the average initial moment vector. Recovering Σ from $M - \mu\mu^T$ gives the covariance. ■

Proposition 4 (Existence convexity):
If an AA existence update uses nonnegative weights that sum to one, then the fused existence probability remains in the convex hull of the incoming existence probabilities.

Proof:
The fused value is a convex combination of values in $[0, 1]$. ■

Proposition 5 (Reference-label invariance):
For every sensor s, the label set of $Y_s(k)$ is a subset of the label set produced by one sensor in $\mathcal{N}_s$. Moreover, if every estimate in $\mathcal{N}_s$ has the same labels and identical matched Gaussian moments, then $\mathcal{B}_s$ returns that estimate unchanged.

Proof:
The operator defines output labels only from the selected reference $X_{\rho_s(k)}(k)$, so it never creates a label outside a neighborhood estimate. If all neighborhood estimates share the same labels and moments, every admissible reference has the same label set, each assignment is the identity up to ties among identical states, and each matched group contains identical moments. The barycenter equations therefore return the same mean and covariance. ■

These properties are structural. The consensus-limit result is conditional on stable matching and stochastic graph weights; the reported implementation uses three finite rounds and recomputes references and assignments. None of the propositions implies that tracking error must decrease for every scene. The empirical section therefore tests the operator against a tuned spatial-KLA AA baseline and a reference-only ablation.

VI. EXPERIMENTAL SETUP

The evaluation uses an eight-sensor formation-tracking scenario with tiered fixed packet-loss probabilities. Each 50-trial paired run uses the same seeds and the same per-trial packet-loss assignments across methods. The loss levels are $\{0, 0.1, 0.2, 0.5\}$, with one sensor at 0, four sensors at 0.1, one sensor at 0.2, and two sensors at 0.5 in each trial. The LMB filters use AA parallel update mode, Metropolis fusion weights, maximum three Gaussian-mixture components, 150 loopy-belief-propagation iterations, and an existence threshold of 0.18.

We compare three arms chosen to isolate the mechanism. The baseline is a tuned spatial-KLA AA implementation with target-wise weight handling, link-quality weighting, existence confidence, and structure-aware spatial decoupling; it tests whether label-space errors remain after scalar fusion weights are routed to the correct consumers. The proposed arm enables neighborhood label-barycenter projection over communication neighborhoods with $H = 3$ iterations; it tests the full label-canonicalization and matched-barycenter hypothesis. The ablation arm uses the same reference selection and label canonicalization but copies the medoid reference states without matched-state barycenters; it tests whether label-set agreement alone explains the gain.

Network disagreement is measured by pairwise output OSPA, localization disagreement, and cardinality dispersion across sensors. Truth-referenced tracking quality is measured by local E-OSPA, RMSE, and cardinality error averaged across sensors. OSPA is used because it consistently combines localization and cardinality errors for multi-object filters [19]. We report mean values, 95% confidence intervals over trials, paired reductions relative to the tuned baseline, wins out of 50 paired trials, and sign-test p values.

With base seed 1, the 50 trial seeds are 2 through 51. Each trial samples one eight-sensor packet-loss vector from the tiered profile and reuses it for all arms. Intervals are mean ± 1.96 standard errors over trial-level means; paired reductions use baseline-minus-candidate deltas. Wins and two-sided sign-test values are computed on nonzero paired deltas.

The numerical tables and Fig. 3 are generated from tracked validation reports rather than edited by hand. The reproducibility package records the source report path, source SHA-256 digest, and generated machine-readable summaries for the N50 AA validation and the contextual GA reference rows. A verifier recomputes network-disagreement statistics from per-trial report rows and runtime statistics from trial logs before the PDF is built.

TABLE I

Fifty-trial paired validation under tiered packet loss. Lower values are better. Values are trial means. The proposed operator improves both network agreement and local tracking quality; the reference-only ablation isolates the part of the gain that comes from label-set canonicalization alone.

Method	Net OSPA	Loc. disag.	Card. disp.	E-OSPA	RMSE	CardErr
Tuned spatial-KLA AA	1.682607	1.472837	0.035350	2.029641	3.682880	0.088000
Neighborhood reference-only	1.030062	0.891585	0.021200	1.911286	3.662955	0.077200
Neighborhood label-barycenter	0.309818	0.216372	0.021200	1.681483	3.449035	0.077200

VII. RESULTS

Table I shows that the proposed operator improves the communication-facing and truth-facing metrics simultaneously. The 95% confidence intervals for network OSPA are $[1.666520, 1.698693]$ for the tuned baseline, $[0.300179, 0.319457]$ for the full method, and $[1.014315, 1.045808]$ for the reference-only ablation. For local E-OSPA, the corresponding intervals are $[2.012605, 2.046677]$, $[1.663594, 1.699373]$, and $[1.893427, 1.929145]$.

Table II places the proposed AA projection beside two GA reference rows generated with the same base seed, trial seeds, and tiered packet-loss profile. Because those rows come from the GA validation path, they are used only as contextual baselines rather than as paired significance-test inputs. In this setting, the neighborhood label-barycenter row is lower than both GA reference rows on all six reported disagreement and tracking metrics, which supports the interpretation that label canonicalization and matched posterior barycenters address a failure mode not removed by the existing GA weighting variants alone.

Table III and Fig. 3 separate label-set effects from state-barycenter effects. The reference-only ablation reduces network disagreement because sensors in a neighborhood move toward a common label set, but its RMSE reduction is only 0.54%, with a confidence interval crossing zero, 27 wins out of 50 paired trials, and no sign-test support. The full operator reaches 6.35% RMSE reduction, with 48 wins out of 50 and $p < 10^{-3}$. The full method also improves local E-OSPA in all 50 paired trials and cardinality error in 46 out of 50 paired trials. These results support the claim that matched posterior barycenters, not label copying alone, drive the spatial tracking gain.

The computational cost is higher than the tuned baseline. Mean filter runtime increases from 62.245 s to 102.347 s, or 1.647 times the tuned baseline, for the 50-trial validation (Table IV). The reference-only ablation costs 1.542 times the baseline, so most of the overhead comes from repeated neighborhood matching and output rewriting rather than from moment computation alone.

TABLE II
Contextual reference rows from the same 50 trial seeds and tiered packet-loss profile. Lower values are better. The GA rows are generated by the tracked GA validation path and are not used in the paired AA sign tests.

Method	Net OSPA	Loc. disag.	Card. disp.	E-OSPA	RMSE	CardErr
GA +structure-aware KLA	1.761137	1.462915	0.084125	2.213284	4.228361	0.203975
GA +FID-FIA refinement	1.847668	1.958378	0.081275	2.179948	4.695098	0.151025
Neighborhood label-barycenter	0.309818	0.216372	0.021200	1.681483	3.449035	0.077200

TABLE III
Paired reductions relative to the tuned spatial-KLA AA baseline. The bracketed interval is the 95% confidence interval of the absolute paired reduction, and p is the two-sided sign-test value on nonzero paired deltas.

Metric	Full reduction	Full wins	Full p	Ref.-only reduction	Ref.-only wins	Ref. p
Network OSPA	81.59% [1.359, 1.386]	50/50	$< 10^{-3}$	38.78% [0.639, 0.666]	50/50	$< 10^{-3}$
Loc. disagreement	85.31% [1.238, 1.275]	50/50	$< 10^{-3}$	39.46% [0.566, 0.596]	50/50	$< 10^{-3}$
Card. dispersion	40.03% [0.012, 0.016]	50/50	$< 10^{-3}$	40.03% [0.012, 0.016]	50/50	$< 10^{-3}$
E-OSPA	17.15% [0.342, 0.355]	50/50	$< 10^{-3}$	5.83% [0.112, 0.125]	50/50	$< 10^{-3}$
RMSE	6.35% [0.201, 0.267]	48/50	$< 10^{-3}$	0.54% [-0.017, 0.056]	27/50	0.672
Card. error	12.27% [0.009, 0.013]	46/50	$< 10^{-3}$	12.27% [0.009, 0.013]	46/50	$< 10^{-3}$

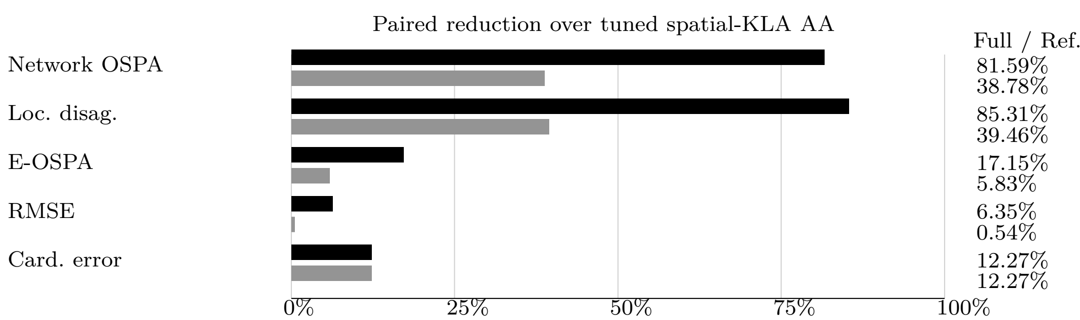

Fig. 3. Paired percentage reductions over the tuned spatial-KLA AA baseline. Black bars show the full label-barycenter operator, and gray bars show the reference-only ablation. The reference-only ablation captures the effect of choosing and copying a consistent local reference label set. The full method gives substantially larger localization and RMSE gains because it also averages matched posterior states.

TABLE IV
Filter runtime over 50 paired trials.

Method	Runtime (s)	Relative
Tuned spatial-KLA AA	62.245 ± 6.444	1.000x
Neighborhood reference-only	95.576 ± 11.190	1.542x
Neighborhood label-barycenter	102.347 ± 11.262	1.647x

VIII. DISCUSSION AND LIMITATIONS

The present operator is an estimate-level projection layer. It is graph-local because each sensor uses only its communication neighborhood, but it is not a full recursive message update inside the LMB filtering loop. The results therefore support label-aligned output fusion under unreliable communication. A recursive implementation would also need lifecycle guards for birth, death, and intermittently matched labels, because unstable matches can turn a useful barycenter into an erroneous track merge.

The method uses Euclidean position matching and equally weighted moment barycenters. These choices keep the operator interpretable and make the reference-only ablation clean, but they do not use covariance quality, label age, or link reliability inside the barycenter itself. A covariance-aware or reliability-weighted variant is a natural extension when local uncertainty differs strongly across sensors. The validation is also concentrated on a tiered packet-loss formation scenario. Broader topologies, partial-field-of-view stress cases, and covariance-consistency diagnostics are needed before claiming robustness across all distributed tracking regimes.

IX. CONCLUSION

This paper reframes a persistent AA-LMB fusion error as a label-space alignment and matched-posterior barycenter problem. The proposed neighborhood label-barycenter operator chooses a robust local reference, canonically matches neighboring tracks to that reference, and computes first-two-moment barycenters for matched states. In 50 paired trials under tiered packet loss, the method improves both network agreement and local tracking metrics, while a reference-only ablation shows that the main spatial gain comes from posterior barycentering rather than label copying alone. The result is a graph-local projection layer that can be embedded into future recursive distributed LMB designs with explicit lifecycle and consistency safeguards.

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